Ashee B Bansaal

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EDUCATION

Cornell University, GPA: 3.846 / 4.0 **Expected Graduation: May 2028**
Bachelor of Science: Computer Science (Honors); Specialization: ML, AI; Minor: Electrical and Computer Engineering
Relevant Coursework: Analysis of Algorithms, Systems for Large-Scale ML, Machine Learning (TA FA26), Signals and Systems, Embedded Systems, Data Science, Object-Oriented Programming, Digital Logic, Data Structures, Virtual Reality
Honors & Awards: Cornell BURE Grant'25 (1 of 40 undergraduates; $7,000); Cornell Summer Research Grant'26 ($6,000)

PROFESSIONAL EXPERIENCE

Carnegie Mellon University, Semantic Signal Lab June 2026 - August 2026
Computer Vision Intern | ICLR' 27
- Developed a gaze-guided visual memory framework for long egocentric video Q&A using VLMs, achieving 53.5% recall.
- Engineered a cross-session identity tracking pipeline using DINOv2 embeddings and Sinkhorn transport across 10+ hours video.
- Devised high-performance GPU embedding deduplicating engine in CUDA/PyTorch, optimizing parallel execution, achieving 100% CPU parity for 3,700+ targets in 283s.

Cornell University, Sci-Fi Lab June 2025 - August 2025
Machine Learning Intern | CHI' 27
- Constructed a PyTorch multimodal Conformer–CTC model fusing 4-channel acoustic and 6-axis IMU data for 35 gesture classes.
- Trained software architecture on 6K sequences using cross-user pretraining, 5-fold personalization, and multitask learning.
- Achieved 98.9% gesture accuracy and 97.4% facial expression macro-F1, outperforming CTC baselines by up to 24.4%.

Cognify: Hands-Free Brain-Computer Interface August 2025 - Present
Founder & Machine Learning Engineer
- Designed SSVEP BCI classifier using Conv-LSTM+attention ensembles, achieving 95.3% accuracy and 40% faster training.
- Configured a multimodal motor-imagery model combining EEGNet and temporal CNNs, improving accuracy to 81.25% (+6%).
- Configured scalable MLOps and containerized deployment using Optuna, Kubernetes, and Docker, reducing iteration time 2x.

Cult.Fit May 2024 - July 2024
App Developer
- Shipped Kotlin/Java Health Connect features for a 150K-MAU Android fitness app, syncing wearables across 6+ devices.
- Built multi-threaded coroutine pipelines with retries and circuit breakers, processing 500K+ daily events at <100ms p95 latency.
- Reduced SQLite latency 60% via WAL optimization; ran Firebase–BigQuery A/B tests across 50K users, driving 40% adoption..

PROJECT & RESEARCH EXPERIENCE

CalibraType: Calibration-Free Adaptive Mobile Keyboard | CHI'27 May 2025 - Present
Cornell University, Sci-Fi Lab
- Built a C++ online probabilistic adaptation model that filters noisy touch events and incrementally updates per-key 2D Gaussian distributions from real-world typing data, reducing input errors by 28%.

Select-or-Sign: AI-powered AR Communication Interface | CHI'27 August 2025 - July 2026
Cornell University, Sci-Fi Lab
- Engineered a 7-task asynchronous real-time AR pipeline using Whisper ASR, GPT-4o-mini, and MediaPipe with <1s latency.
- Deployed a personalized TFLite gesture model trained on 5K+ poses, achieving 95%+ accuracy and <5 min calibration.

Spatial Moves as Metacognitive Regulation January 2025 - July 2025
Cornell University, DAIL lab
- Built an interactive React/TypeScript/D3 skill-roadmap interface for real-time visualization of personalized learning graphs.
- Implemented FAISS/Pinecone vector search in Python/Node.js backend to generate 100+ adaptive learning nodes in Neo4j.

LEADERSHIP EXPERIENCE

Cornell University Extended Reality (XR) Project Team January 2025 - Present
BCI sub-team Lead
- Implemented a real-time 256 Hz EEG inference pipeline using FFT spectral features, achieving 82% accuracy across 20+ users.
- Led a 6-member team through GitHub Actions CI, code reviews, and sprint planning, reducing integration bugs by 40%.

Cornell IEEE January 2025 - January 2026
Director of Corporate Relations
- Secured partnerships with ~150 companies, including Marvell Technology, to organize Cornell IEEE corporate events.

TECHNICAL SKILLS

- **Languages :** Python, C/C++, Go, Java, JavaScript, TypeScript, Swift, SQL, Solidity, Rust, Bash/ Linux, R, MATLAB
- **AI/ ML:** PyTorch, TensorFlow, LangChain, CUDA, Scikit-learn, Keras, Hugging Face, LangGraph, OpenCV, CUDA, Triton
- **Infra & Tools**: AWS, GCP, Docker, Kubernetes, Git, Redis, PostgreSQL, Linux, ROS, Firebase, Pytest, Claude Code, Cursor